

ISOSAARI, Finland

WMO: 029880

Lat: 60.100N

Lon: 25.051E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (EUE)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.2	-14.9	-20.6	0.6	-18.1	-17.0	0.8	-14.6	18.1	-0.5	15.8	-1.2	5.2	40	0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.6	23.0	19.0	21.8	18.5	20.7	17.7	19.9	22.1	19.0	21.2	18.2	20.2	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	21.6	18.1	13.0	20.5	17.2	12.3	19.9	57.0	22.2	54.1	21.3	51.3	20.2	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	13.8	12.3	-19.5	24.6	3.0	1.7	-21.7	25.8	-23.5	26.8	-25.2	27.7	-27.4	29.0
			-20.4	20.9	3.4	1.7	-22.8	22.1	-24.8	23.0	-26.7	24.0	-29.1	25.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-2.3	-4.4	-1.9	3.1	8.0	13.6	17.3	17.1	12.5	7.4	2.6	-0.2	(6)
(7)		DBStd	8.20	4.83	5.57	4.05	2.36	2.63	2.63	2.63	2.44	2.49	3.55	3.54	4.48	(7)
(8)		HDD10.0	2076	382	404	369	208	73	2	0	0	6	94	222	315	(8)
(9)		HDD18.3	4491	641	638	627	458	321	142	51	54	176	338	472	573	(9)
(10)		CDD10.0	663	0	0	0	0	11	111	228	219	80	14	0	0	(10)
(11)		CDD18.3	36	0	0	0	0	0	1	20	15	0	0	0	0	(11)
(12)		CDH23.3	24	0	0	0	0	0	0	19	5	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.0	7.0	6.5	5.6	5.3	5.1	4.8	5.0	5.5	5.9	6.6	7.3	7.2	(14)
(15)	Precipitation	PrecAvg	596	49	33	32	33	34	51	57	70	55	72	66	48	(15)
(16)		PrecMax	739	85	101	57	68	74	102	140	160	129	180	172	88	(16)
(17)		PrecMin	367	1	0	4	8	6	19	2	3	14	29	4	10	(17)
(18)		PrecStd	97	21	24	14	17	20	23	41	41	31	34	36	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	3.6	7.0	11.0	17.0	21.8	25.3	24.3	19.0	14.4	9.7	7.7	(19)
(20)			MCWB	4.6	2.5	4.0	7.1	11.9	17.3	20.5	20.3	15.8	12.8	8.7	6.9	(20)
(21)		2%	DB	4.2	2.7	4.5	8.8	14.7	20.0	23.6	22.6	17.7	13.6	9.2	7.0	(21)
(22)			MCWB	3.6	2.1	2.2	6.0	10.8	16.3	19.3	19.1	15.4	12.3	8.3	6.3	(22)
(23)		5%	DB	3.4	2.1	3.1	7.3	12.9	18.7	22.3	21.6	16.9	13.0	8.6	6.3	(23)
(24)			MCWB	2.7	1.6	1.7	5.2	9.4	15.1	18.8	18.4	14.9	12.0	7.8	5.6	(24)
(25)		10%	DB	2.6	1.6	2.1	6.2	11.6	17.5	21.2	20.6	16.0	12.2	7.5	5.1	(25)
(26)			MCWB	1.9	1.0	1.1	4.4	8.7	14.5	18.1	17.8	14.3	11.2	6.6	4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	3.0	4.2	8.0	13.8	18.6	22.2	20.7	16.7	13.3	9.2	7.2	(27)
(28)			MCDWB	5.1	3.3	6.8	10.2	15.1	20.6	24.2	22.8	18.0	13.8	9.5	7.3	(28)
(29)		2%	WB	3.7	2.2	2.7	6.5	11.2	17.0	20.2	19.8	16.0	12.7	8.6	6.5	(29)
(30)			MCDWB	4.1	2.4	3.8	8.1	13.4	19.2	22.3	22.0	17.3	13.3	9.1	6.9	(30)
(31)		5%	WB	2.9	1.7	1.9	5.5	9.9	15.8	19.4	19.0	15.3	12.0	7.9	5.6	(31)
(32)			MCDWB	3.3	2.0	2.7	7.0	12.4	17.8	21.6	21.0	16.6	12.9	8.6	6.1	(32)
(33)		10%	WB	2.0	1.2	1.3	4.7	9.0	14.8	18.5	18.2	14.5	11.3	6.7	4.4	(33)
(34)			MCDWB	2.4	1.5	1.9	6.0	11.2	16.9	20.7	20.1	15.8	12.2	7.3	5.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.4	3.9	4.2	3.4	4.0	3.8	3.6	3.4	3.2	2.6	2.8	3.3	(35)
(36)			MCDBR	2.8	2.5	4.0	5.2	5.9	5.4	4.7	4.0	3.7	2.2	2.7	2.0	(36)
(37)			MCWBR	3.1	2.7	2.6	3.3	3.5	3.2	3.1	2.5	2.8	2.1	3.1	2.3	(37)
(38)		5% WB	MCDWB	2.9	2.4	3.4	4.9	5.1	4.6	3.7	3.4	3.3	2.1	2.4	1.8	(38)
(39)			MCWBR	3.2	2.7	2.6	3.4	3.6	3.4	2.8	2.5	2.7	2.1	3.0	2.3	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.280	0.312	0.334	0.361	0.359	0.357	0.379	0.373	0.346	0.334	0.308	0.265	(40)
(41)		taud		2.146	2.194	2.246	2.332	2.410	2.480	2.441	2.456	2.520	2.396	2.218	2.096	(41)
(42)		Ebn at Noon		476	672	787	837	868	876	844	820	785	655	439	338	(42)
(43)		Edh at Noon		43	72	94	104	103	98	100	92	73	60	41	30	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.25	0.82	2.31	3.85	5.28	5.49	5.29	4.10	2.50	1.09	0.30	0.15	(44)
(45)		RadStd		0.04	0.15	0.30	0.40	0.57	0.42	0.48	0.42	0.32	0.17	0.05	0.02	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	-140	N/A	N/A	N/A

Nomenclature: See separate page